Sales Data Cleaning & Analysis

Fresher Data Analytics Assignment | Dataset: 2,823 sales line items | Period: Jan 2003 – May 2005

Executive summary

The dataset contains **2,823** transaction-line records. Gross recorded sales are **$10,032,628.85**. For the primary operating KPIs, cancelled rows are excluded, giving **$9,838,141.37** across **303** distinct orders and an average order value of **$32,469.11**. The analysis covers 2003-01 to 2005-05; 2005 is a partial year.

KPI	Result
Gross recorded sales	$10,032,628.85
Operating sales (excl. Cancelled)	$9,838,141.37
Average order value	$32,469.11
Distinct operating orders	303
Distinct customers	92
Units ordered	97,029

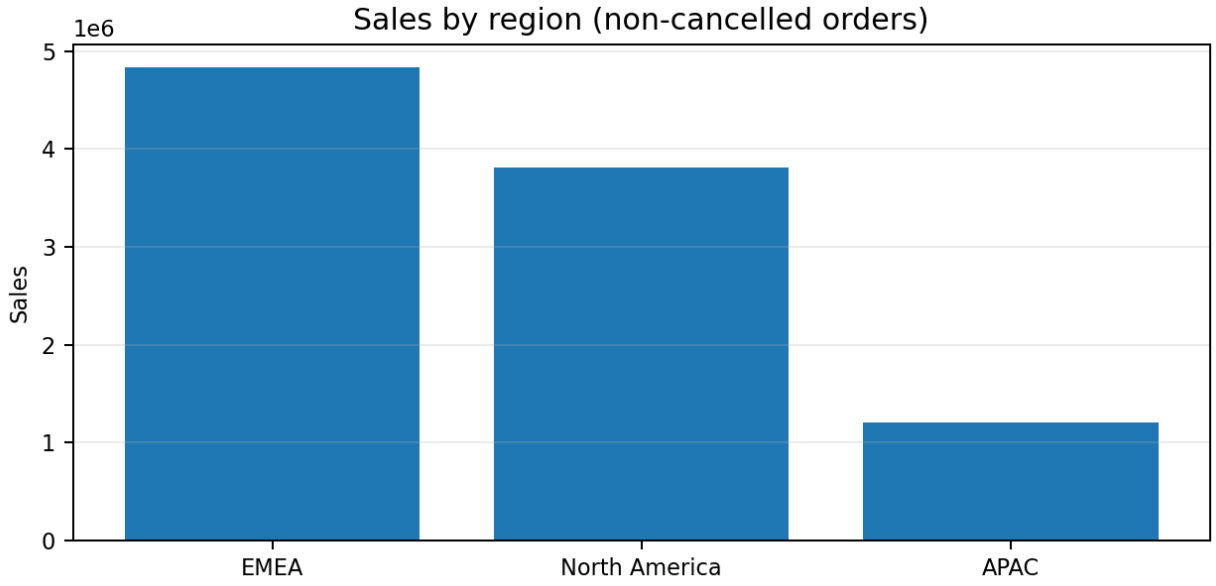
Data preparation & quality findings

The source file was read using Windows-1252 encoding because it contains non-UTF-8 characters. Text fields were trimmed, whitespace normalised, numeric measures converted to numeric types, and ORDERDATE parsed as a date. Optional missing address and territory fields were labelled Unknown. A new REGION field was derived consistently from COUNTRY; the original TERRITORY field was retained for traceability. No exact duplicate rows, duplicate order-line keys, invalid dates, missing critical sales/date/product fields, or negative sales values were detected.

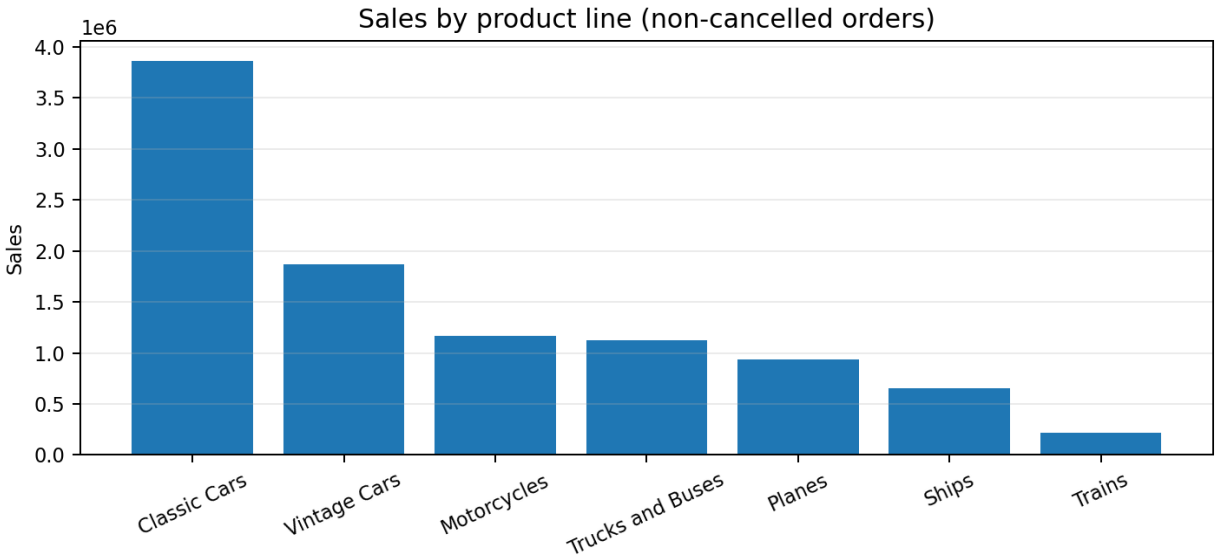
Quality check	Count	Treatment
Exact duplicate rows	0	No exact duplicate records detected.
Duplicate order-line keys	0	No repeated ORDERNUMBER + ORDERLINENUMBER keys detected.
Missing ADDRESSLINE2	2521	Optional address detail; labelled Unknown in cleaned file.
Missing STATE	1486	Geographic detail; labelled Unknown in cleaned file.
Missing POSTALCODE	76	Postal detail; labelled Unknown in cleaned file.
Missing TERRITORY	1074	1074 blanks; REGION derived consistently from COUNTRY.
Missing critical sales/date/product fields	0	No missing values found in these critical fields.
Invalid/unparsed order dates	0	Dates parsed to datetime; no invalid dates found.
Negative sales values	0	No negative sales values detected.
Cancelled rows excluded from operating KPI	60	Cancelled line items retained in cleaned CSV, excluded from operating sales/AOV/trend.

Sales insights

Region: EMEA leads with $4,830,142.59, followed by North America ($3,806,703.73) and APAC ($1,201,295.05).

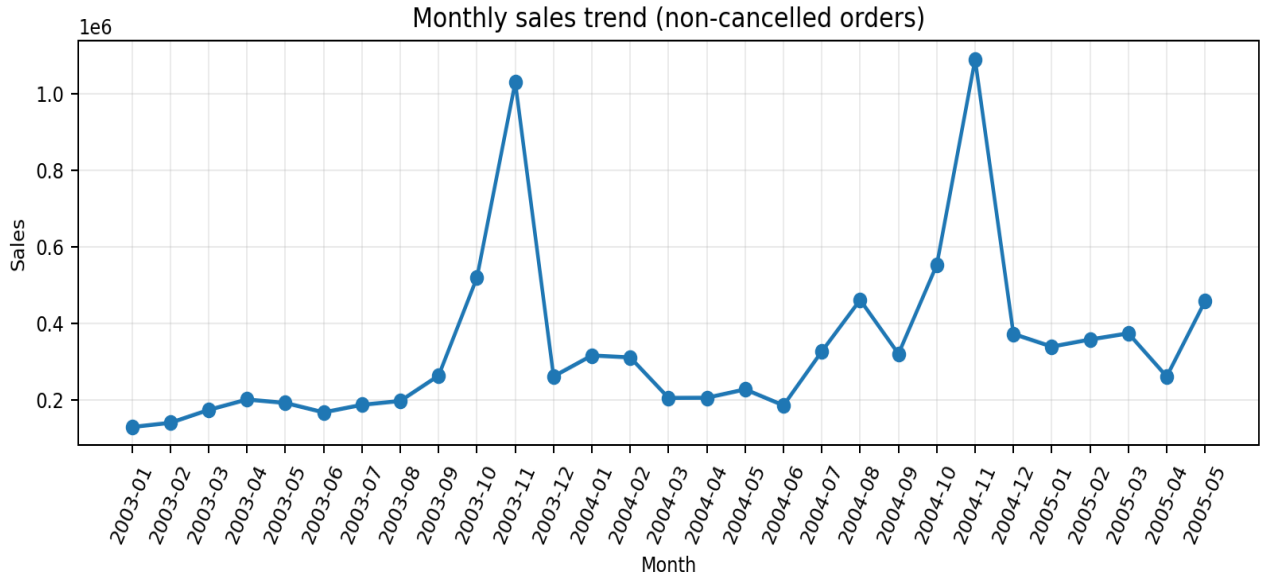


Product: Classic Cars is the top product line at $3,860,372.85; Trains is the lowest at $221,161.05.



Monthly sales trend

Monthly totals are based on non-cancelled line items. November is the highest sales month in both 2003 and 2004. The dataset ends in May 2005, so 2005 should not be compared with complete years without adjusting for the shorter period.



Five actionable recommendations

1. Prioritise the strongest product line — Classic Cars is the largest product-line contributor. Protect stock availability, review margins, and test bundles or accessories to increase basket size.

2. Defend core regions and grow APAC — EMEA and North America generate most operating sales. Strengthen key-account retention there while testing targeted campaigns in APAC.

3. Prepare earlier for Q4 demand — November is the strongest month in both complete years. Align inventory, staffing, and marketing campaigns ahead of October–November.

4. Investigate cancellations — Cancelled rows account for $194,487.48 in recorded line value. Track cancellation reasons and review order confirmation, credit checks, and fulfilment delays.

5. Improve data capture at source — Territory is missing on 1,074 rows, while state and postal code are also frequently blank. Make required fields conditional by country and maintain a controlled country-to-region mapping.

Method notes & limitations

AOV is calculated as total non-cancelled sales divided by distinct non-cancelled ORDERNUMBER values (order-level AOV, not average line value). Cancelled records remain in the cleaned dataset but are excluded from operating sales, AOV, and trend charts. Other statuses (Shipped, Resolved, On Hold, In Process, Disputed) are retained; therefore operating sales is a practical non-cancelled sales measure, not audited recognised revenue. Region is a derived analytical grouping from COUNTRY: USA/Canada = North America; Australia/Japan/Philippines/Singapore = APAC; listed European countries = EMEA. Currency symbol is shown as $ because the source does not provide a separate currency field.